

```

s1
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
*** Stopping 0 controllers

*** Stopping 2 links
..
*** Stopping 1 switches
..
*** Stopping 2 hosts
h1 h2
*** Done
completed in 0.057 seconds

```

Figure 13b: Topology creation and validation tests at mining

```

mininet> h1 ping -c10 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=40.5 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=40.3 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=40.3 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=40.3 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=40.3 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=40.2 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=40.3 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=40.3 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=40.3 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=40.3 ms

--- 10.0.0.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9013ms
rtt min/avg/max/ndev = 40.251/40.357/40.574/0.234 ms

```

Figure 15: Delay induced ping

`xterm` is the command used to initiate an external and separate terminal. The command opens an xterm window, which is specific to a node in the network. Several functionalities specific to a node can be implemented on this window. In the example shown in Figure 14, the `h1` xterm window is

```

root@hp-HP-431-Notebook-PC: ~
root@hp-HP-431-Notebook-PC:~# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.869 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.102 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.100 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.100 ms

```

Figure 14: Testing the availability of host `h2` from host `h1`

used to ping `h2` (10.0.0.2).

- The command to provide a delay of 10ms and set a bandwidth of 10Mbps in all the links is:

```
#sudo mn -link tc,bw=10,delay=10ms
```

Notice that `h1` pinging `h2` takes a total time of 40ms (approximately), since it has to travel through the `s1-h1` link. 

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